

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

**CSA1506**

**CLOUD COMPUTING AND BIG DATA ANALYTICS**

**ASSIGNMENT REPORT**

**PROPERTY BUYING AND RENTAL MANAGEMENT SYSTEM  
FOR CHENNAI USING ZOHO CREATOR (SaaS)  
WITH BIG DATA ANALYTICS**

Submitted by  
**Pugazharasan K**  
**192511341**

Faculty  
**Dr. MD BOOMIJA**

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## **CERTIFICATE**

This is to certify that PUGAZHARASAN K (192511341) has completed the assignment report entitled “Property Buying and Rental Management System for Chennai using Zoho Creator (SaaS) with Big Data Analytics” as part of the requirements of CSA1506 – Cloud Computing and Big Data Analytics under the guidance of Dr. MD BOOMIJA.

The work presented in this report covers the design of a cloud-based Software as a Service application using Zoho Creator, property data management, rental agreement information, cloud workflows and reports, and a Big Data Analytics component using a sample Chennai property dataset and MapReduce-based processing.

Faculty / Guide: Dr. MD BOOMIJA

Signature: Pugazharasan K

Date: 02/09/2026

## **DECLARATION**

I, PUGAZHARASAN K (192511341), hereby declare that this assignment report titled “Property Buying and Rental Management System for Chennai using Zoho Creator (SaaS) with Big Data Analytics” is prepared for the course CSA1506 – Cloud Computing and Big Data Analytics.

The report documents the proposed system design, Zoho Creator implementation, data model, workflows, reports, Big Data problem analysis, MapReduce methodology, testing approach, results, limitations and learning outcomes. External tools, documentation and reference material used for understanding the concepts are acknowledged in the References section.

Student Signature: Pugazharasan K

## **ACKNOWLEDGEMENT**

I express my sincere gratitude to Dr. MD BOOMIJA for the guidance and support provided during the preparation of this assignment. The guidance helped in understanding cloud service models, SaaS application development, data modelling, workflows, reporting and Big Data Analytics concepts.

I also acknowledge the learning resources and software platforms used during the work, especially Zoho Creator for developing the cloud-based application and the tools used to prepare and analyse the sample property dataset.

Finally, I thank the Department of Computer Science and Engineering for providing the academic environment and requirements that supported the completion of this assignment.

## ABSTRACT

The Property Buying and Rental Management System for Chennai is designed as a cloud-based Software as a Service (SaaS) application using Zoho Creator. The system addresses the difficulties involved in managing property buying and rental information through disconnected or manual processes. It provides a centralized application for maintaining property listings, recording buyer and tenant information, supporting Buy/Rent selection, distinguishing Residential and Commercial properties, searching by Chennai locality and budget, recording enquiries and managing rental agreement information.

The proposed application uses structured forms, field validation, workflows, reports and dashboards to improve data consistency and accessibility. The property data model includes Property ID, listing type, category, property type, title, owner/agent details, Chennai locality, address, landmark, BHK/rooms, area, furnishing, amenities, availability, budget, property value, monthly rent, maintenance, advance/deposit, agreement duration and property status. The submitted Zoho Creator implementation evidence includes a Property Buying and Selling form and report/dashboard components.

The project also examines Big Data challenges associated with a growing property dataset, including volume, velocity, variety, veracity, duplicate listings, location data, privacy and scalability. A simple MapReduce methodology is proposed for area-wise listing counts, property-type demand and average price/rent analysis. Testing is defined for functional validation, data validation, workflows, reports and analytics. The project demonstrates how SaaS, cloud collaboration, data management and Big Data processing concepts can be combined in a practical real-estate application.

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## 1. INTRODUCTION

Property buying and rental activities involve several pieces of information such as property type, location, size, price, rent, owner details, availability and agreement terms. In a large city such as Chennai, property information can change frequently and users may need to compare listings across different localities. A centralized cloud application can make this information easier to organize and access.

The proposed Property Buying and Rental Management System provides a structured method for storing and managing property listings. Users can identify whether they are interested in buying or renting and can distinguish Residential and Commercial properties. Property information can then be searched or reported based on locality, budget, property type and availability.

Zoho Creator is used as the SaaS platform for implementing the application. The cloud-based approach reduces the need for maintaining a separate application server and enables access through a web interface. Forms are used for data collection, while reports and dashboards provide organized views of the stored information.

The project additionally introduces Big Data Analytics concepts. A property platform can generate a growing stream of listing, enquiry, visit, search and transaction information. Analysing this data can reveal locality demand, property-type demand and price/rent patterns. MapReduce is used as a simple distributed-data-processing model for demonstrating such analysis.

## 2. PROBLEM STATEMENT AND PROBLEM FORMULATION

### 2.1 Problem Statement

Design, implement and evaluate a Property Buying and Rental Management System for Chennai city using Zoho Creator as a cloud-based SaaS application. The system should support property listing, Buy/Rent selection, Residential/Commercial classification, property search, enquiries, visits, shortlisting, buying or rental requests, owner/agent listing updates, property status management and rental agreement information.

The system should maintain essential property fields and provide forms, validations, workflows, role-based access, reports, dashboards, alerts and cloud collaboration. The project should also prepare a sample Chennai property dataset, identify Big Data challenges and demonstrate a MapReduce application for property analytics.

### 2.2 Problem Formulation

Inputs include property details, owner/agent information, buyer/tenant information, enquiries, visit requests, rental agreement details and property status updates. Processing includes validation, classification, searching, filtering, workflow-based status changes and analytics. Outputs include organized property reports, dashboards, agreement information and analytical summaries.

| Component | Description |
|-----------|-------------|
|-----------|-------------|

|            |                                                                                     |
|------------|-------------------------------------------------------------------------------------|
| Input      | Property, owner/agent, buyer/tenant, enquiry, visit and agreement information       |
| Processing | Validation, classification, search/filter, workflow, reporting and analytics        |
| Output     | Property listings, status reports, dashboard views, agreement records and analytics |
| Users      | Buyer, tenant, owner/agent and administrator                                        |

### 3. OBJECTIVES AND EXPECTED OUTCOMES

#### 3.1 Objectives

1. Develop a cloud-based property management application using Zoho Creator as SaaS.
2. Provide Buy and Rent selection for property transactions.
3. Support Residential and Commercial property categories.
4. Maintain property details including locality, area, BHK/rooms, furnishing, value, rent and availability.
5. Support Chennai-specific location and budget-based property search.
6. Maintain owner/agent and buyer/tenant information.
7. Provide property enquiry and visit-request information.
8. Maintain Rental Agreement details such as rent, advance, duration, terms and renewal information.
9. Apply validation and workflow rules to improve data quality and property-status management.
10. Use reports, dashboards and a sample Big Data dataset to demonstrate property analytics.

#### 3.2 Expected Outcomes

- A functional cloud application structure for property buying and rental management.
- Structured property and rental agreement data models.
- Validated forms and workflow-driven property status management.
- Searchable reports and dashboard views.
- A clean sample Chennai property dataset suitable for analytics.
- MapReduce-based analytical output such as area-wise listing counts and property-type demand.
- Testing evidence and documented observations.
- Demonstration of CO1–CO5 concepts and SDG relevance.



## 4. REQUIREMENTS, CONSTRAINTS AND ASSUMPTIONS

### 4.1 Functional Requirements

- User registration/login and access control.
- Property registration and listing update.
- Buy/Rent selection.
- Residential/Commercial classification.
- Property type selection such as House, Flat, Villa, Shop, Office and Land.
- Search/filter by Chennai locality, budget, property type and availability.
- Buyer and tenant information management.
- Property enquiry and visit scheduling information.
- Shortlisting and buying/rental request information.
- Rental Agreement record creation and maintenance.
- Property status management using Available, Booked, Sold and Rented states.
- Reports and dashboards for property analysis.

### 4.2 Non-Functional Requirements

- Cloud accessibility through a supported browser.
- Simple and understandable form design.
- Consistent field names and data types.
- Validation for mandatory and logically related fields.
- Scalability for increasing property and transaction records.
- Data privacy and controlled access to owner, tenant and contact information.
- Reliable reporting and repeatable analytics processing.

### 4.3 Constraints

- The academic implementation is a prototype and may not represent an industrial-scale real-estate platform.
- Real-time external property feeds are outside the scope of the classroom implementation.
- Availability of some advanced platform features may depend on the Zoho Creator account/plan.
- Analytics results depend on the quality and size of the sample dataset.

### 4.4 Assumptions

- Property records entered into the application are assumed to be supplied by authorized users.
- Chennai locality names are entered consistently for analysis.
- Buy/Rent and Residential/Commercial selections are mutually meaningful for each listing.
- Rental agreement information is entered only when a rental process requires it.
- Analytics is performed on a prepared dataset exported or constructed from application records.

## 5. APPLICATION OF COURSE CONCEPTS

| CO  | Course Concept                          | Application in Project                                                                                                   |
|-----|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| CO1 | Cloud service models                    | Zoho Creator is used as SaaS; IaaS, PaaS and SaaS are compared.                                                          |
| CO2 | Virtualization and cloud infrastructure | Conceptual virtualization layer explains how compute, storage and network resources support cloud hosting.               |
| CO3 | Cloud access and collaboration          | Browser-based access, role-based usage, shared data, workflows, reports and dashboards.                                  |
| CO4 | Big Data problems                       | Property data growth is analysed for volume, velocity, variety, veracity, duplicates, location, privacy and scalability. |
| CO5 | MapReduce                               | Mapper/reducer logic is used to aggregate locality counts, property-type demand and price/rent metrics.                  |

The project therefore connects the application development portion with the analytical portion of the course. The Zoho Creator application demonstrates cloud service usage, while the data preparation and MapReduce component demonstrates how a property platform can derive useful information from growing datasets.

## 6. CLOUD SERVICE MODEL – SaaS

### 6.1 Cloud Service Models

| Model | Main Responsibility                                                                                     | Typical Use                                 |
|-------|---------------------------------------------------------------------------------------------------------|---------------------------------------------|
| IaaS  | Provider supplies virtualized compute, storage and networking; user manages more of the software stack. | Virtual machines and infrastructure hosting |
| PaaS  | Provider manages infrastructure and runtime; user develops/deploys applications.                        | Application development platforms           |
| SaaS  | Provider delivers the complete application/service through the cloud.                                   | Browser-based business applications         |

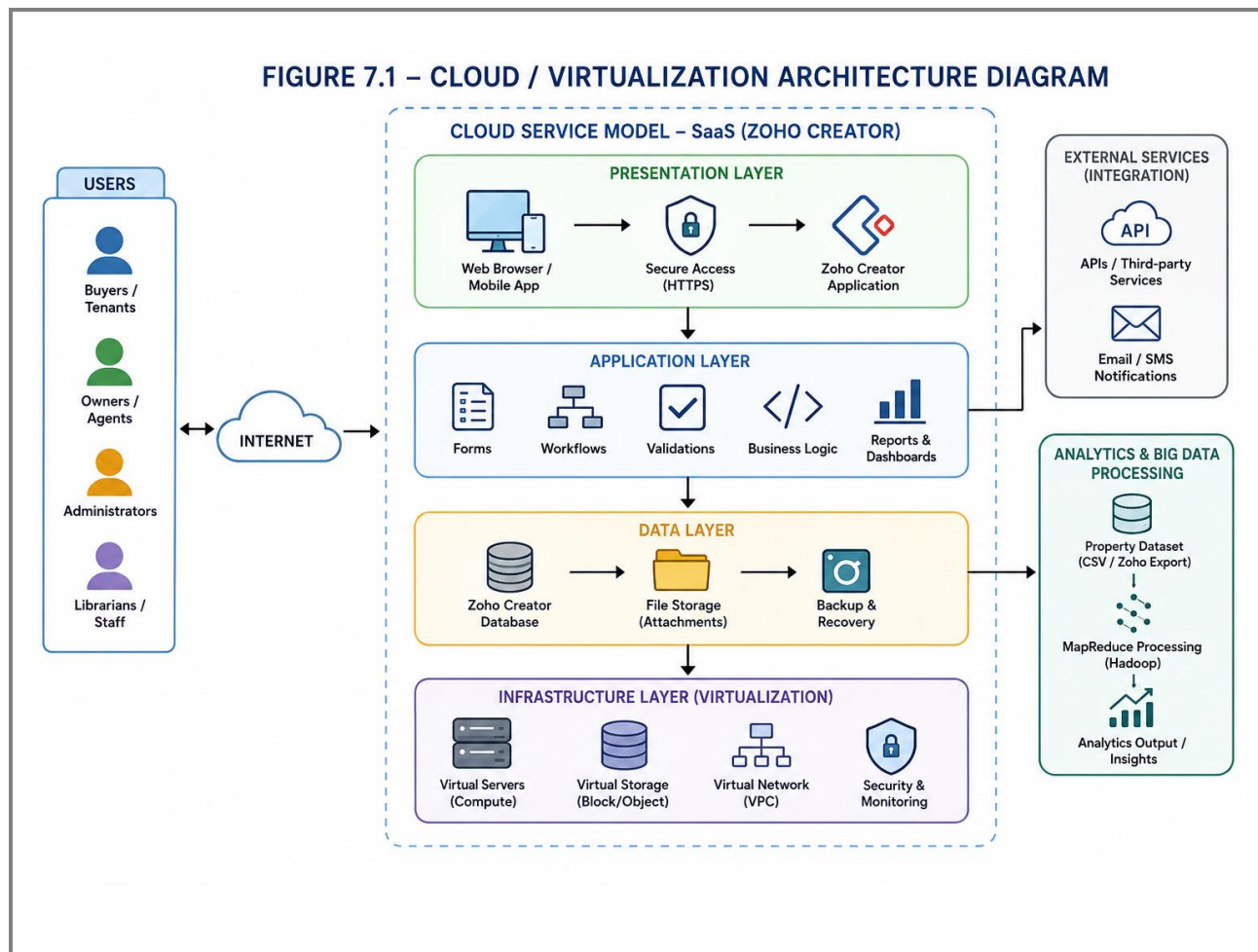
### 6.2 Why Zoho Creator is Suitable

- Provides cloud-based application development through forms, workflows and reports.
- Reduces the need to build and maintain a complete backend stack for the academic prototype.
- Supports structured data collection and application-level automation.
- Allows application access through a web interface.
- Provides reporting and dashboard capabilities appropriate for a property management prototype.

In this project, Zoho Creator is treated as the SaaS application platform. The student interacts mainly with the application configuration and data model rather than managing the underlying physical servers.

## 7. VIRTUALIZATION AND CLOUD INFRASTRUCTURE DESIGN

Cloud platforms depend on abstraction of physical computing resources. Virtualization allows physical compute, memory, storage and networking resources to be represented as logical resources that can be allocated and managed efficiently. For this project, virtualization is presented as the conceptual infrastructure layer beneath the SaaS application.



Conceptual flow: User Browser → Cloud Access → SaaS Application Layer → Zoho Creator Forms/Workflows → Cloud Data Store → Reports/Dashboard → Analytics Dataset → MapReduce Processing.

### 7.1 Infrastructure Components

| Layer               | Role                                                    |
|---------------------|---------------------------------------------------------|
| User Access         | Browser/mobile/desktop access to the application        |
| SaaS Application    | Zoho Creator application and configured business logic  |
| Virtualized Compute | Abstracted compute resources used by the cloud provider |

|              |                                                                  |
|--------------|------------------------------------------------------------------|
| Storage/Data | Application records and prepared analytics dataset               |
| Network      | Cloud connectivity between users and hosted application services |
| Analytics    | Dataset processing and MapReduce-based aggregation               |

## 8. SYSTEM DESIGN AND PROPOSED SOLUTION

### 8.1 Layered Architecture

| Layer             | Functions                                                                        |
|-------------------|----------------------------------------------------------------------------------|
| Access Layer      | Login/access, browser-based application access                                   |
| Application Layer | Property registration, Buy/Rent, categories, enquiry and agreement functions     |
| Data Layer        | Property, user, enquiry, visit and rental agreement records                      |
| Workflow Layer    | Validation, availability/status updates, agreement-related processing and alerts |
| Reporting Layer   | Searchable reports, filtered listings and dashboard views                        |
| Analytics Layer   | Prepared Chennai dataset, MapReduce aggregation and analytical interpretation    |

The screenshot displays the Zoho Creator form builder interface. The browser address bar shows the URL: `creator.zoho.in/appbuilder/19251175.simats_saveetha/property-buying-and-rental/formbuilder/PROPERTY_BUYING_AND_RENTAL/edit`. The form is titled "PROPERTY BUYING AND RENTAL".

**Basic Fields:**

- Name
- Email
- Address
- Phone
- Single Line
- Multi Line
- Number
- Date
- Time
- Drop Down

**Form Fields:**

- TYPE (Selected)
- PROPERTY TYPE
- LOCALITY
- AREA(sq.ft)
- BHK/ROOMS
- BUDGET RANGE
- PROPERTY VALUE
- RENTAL AGREEMENT
- DURATION
- FURNISHING STATUS

**Field Properties:**

- Field name: TYPE
- Field link name: TYPE\_field
- View Field References
- Choices:
  - BUY
  - RENT
  - COMMERCIAL
- Alphabetical Order
- Allow Other Choice

The bottom of the interface shows a "Trial has expired" message with an "Upgrade" button and a "Help" button.

### 8.2 User Roles

| Role | Main Activities |
|------|-----------------|
|------|-----------------|

|               |                                                                                           |
|---------------|-------------------------------------------------------------------------------------------|
| Buyer         | Search properties, view details, submit enquiries, shortlist and initiate buying request. |
| Tenant        | Search rental properties, submit enquiries and provide rental agreement information.      |
| Owner/Agent   | Add/update listings, maintain availability and respond to enquiries.                      |
| Administrator | Manage application data, validation, reports, dashboard and access controls.              |

## 9. PROPERTY DATA MODEL AND FIELD SPECIFICATION

The property record is the central entity of the system. It stores the attributes required to describe a property, classify its transaction type and support search, reporting and analytics.

| Field               | Type / Example                              | Purpose                            |
|---------------------|---------------------------------------------|------------------------------------|
| Property ID         | Text                                        | Unique property identifier         |
| Listing Type        | BUY / RENT                                  | Transaction mode                   |
| Property Category   | RESIDENTIAL / COMMERCIAL                    | Main classification                |
| Property Type       | House / Flat / Villa / Shop / Office / Land | Detailed property type             |
| Property Title      | Text                                        | Human-readable listing title       |
| Owner/Agent Name    | Text                                        | Listing contact person             |
| Contact             | Phone                                       | Owner/agent contact                |
| Area / Locality     | Text                                        | Chennai locality                   |
| Address             | Multi-line text                             | Full property address              |
| Landmark            | Text                                        | Nearby landmark                    |
| BHK / Rooms         | Number/Text                                 | Room configuration                 |
| Built-up Area       | Number                                      | Area in sq.ft                      |
| Carpet Area         | Number                                      | Usable area in sq.ft               |
| Furnishing          | Choice                                      | Furnished/semi/unfurnished         |
| Amenities           | Multi-line / choices                        | Facilities available               |
| Availability        | Choice                                      | Available / unavailable etc.       |
| Budget Range        | Text/Choice                                 | User-facing budget classification  |
| Property Value      | Currency/Number                             | Sale value                         |
| Monthly Rent        | Currency/Number                             | Rental amount                      |
| Maintenance Charge  | Currency/Number                             | Recurring maintenance amount       |
| Advance / Deposit   | Currency/Number                             | Rental advance/deposit             |
| Agreement Duration  | Number/Text                                 | Rental duration                    |
| Agreement Start/End | Date                                        | Agreement period                   |
| Registration Status | Choice                                      | Registration state                 |
| Property Status     | Choice                                      | Available / Booked / Sold / Rented |

## 10. RENTAL AGREEMENT DATA MODEL

The Rental Agreement entity stores the information required after a rental request proceeds toward an agreement. The agreement references the related Property ID so that rental information can be connected to the corresponding property.

| Field        | Purpose                     |
|--------------|-----------------------------|
| Agreement ID | Unique agreement identifier |

|                   |                                      |
|-------------------|--------------------------------------|
| Property ID       | Links agreement to property          |
| Owner Name        | Identifies property owner            |
| Tenant Name       | Identifies tenant                    |
| Monthly Rent      | Agreed rental amount                 |
| Advance / Deposit | Initial amount paid or agreed        |
| Start Date        | Agreement start date                 |
| End Date          | Agreement end date                   |
| Duration          | Agreement period                     |
| Terms             | Important agreement conditions       |
| Renewal Details   | Renewal/extension information        |
| Agreement Status  | Draft / Active / Completed / Renewed |

A rental agreement should only be created for a rental listing and should contain logically consistent dates and financial fields. The Property Status can be changed to Rented after the rental process is completed.

The screenshot displays a web application interface for 'PROPERTY BUYING AND RENTAL'. The main content area shows a report titled 'PROPERTY BUYING AND RENTAL Report' with a table of properties. The table has columns: TYPE, PROPERTY TYPE, LOCALITY, AREA(sq.ft), BHK/ROOMS, BUDGET RANGE, and PROPERTY VALUE. There are 7 rows of data, all with a value of ₹ 1,50,00,000.00. A sidebar on the left contains navigation links: PROPERTY BUYING AND RENTAL, PROPERTY BUYING AND RENTAL Report, and DASHBOARD. Below the sidebar is a user profile card for LOGESHWARAN R with email 192511175.simats@saveetha.com and options for My Account, Home, and Sign out. The bottom of the page shows 'Showing 7 of 7'.

| TYPE       | PROPERTY TYPE     | LOCALITY    | AREA(sq.ft) | BHK/ROOMS | BUDGET RANGE | PROPERTY VALUE   |
|------------|-------------------|-------------|-------------|-----------|--------------|------------------|
| RENT       | WARE HOUSE        | TAMBARAM    | 1036        | 3BHK      | 50-75L       | ₹ 1,50,00,000.00 |
| RENT       | WARE HOUSE        | KILAMBAKKAM | 3023        | 4BHK      | 1Cr+         | ₹ 1,50,00,000.00 |
| RENT       | INDEPENDENT HOUSE | ADAYAR      | 4567        | 3BHK      | 75L-1Cr      | ₹ 1,50,00,000.00 |
| COMMERCIAL | APARTMENT         | VELACHERY   | 6253        |           | 50-75L       | ₹ 1,50,00,000.00 |
| BUY        | APARTMENT         | VELACHERY   |             |           | UNDER 25L    | ₹ 1,50,00,000.00 |
| COMMERCIAL | INDEPENDENT HOUSE | OMR         | 625651      |           | 25-50L       | ₹ 1,50,00,000.00 |
| BUY        | VILLA             | OMR         | 12000       |           | 1Cr+         | ₹ 1,50,00,000.00 |

## 11. ZOHO CREATOR IMPLEMENTATION

### 11.1 Implementation Procedure

11. Open Zoho Creator and create/sign in to the required account.
12. Create a new application for the Property Buying and Rental Management System.
13. Create the main Property form and define the required property fields.
14. Configure choice fields for transaction and property classification.
15. Create supporting forms for buyer/tenant, enquiries, visits and rental agreements as required.
16. Configure field properties, mandatory fields and validation rules.
17. Create workflows for property status and transaction-related events.
18. Create reports for sale, rental, category, locality and agreement views.
19. Create a dashboard to present important property information.
20. Test forms, workflows, reports and analytics using sample Chennai records.

The screenshot displays the Zoho Creator web interface for a form titled 'PROPERTY BUYING AND RENTAL'. The interface includes a top navigation bar with tabs for 'Design', 'Workflow', and 'Settings'. A left sidebar shows a menu with options like 'PROPERTY BUYING AND RENTAL', 'PROPERTY BUYING AND RENTAL Report', and 'DASHBOARD'. The main workspace shows the form layout with the following fields:

- TYPE: Radio buttons for BUY, RENT, and COMMERCIAL.
- PROPERTY TYPE: A dropdown menu with '-Select-'.
- LOCALITY: A dropdown menu with '-Select-'.
- AREA(sq.ft): A text input field with a placeholder '#####'.
- BHK/ROOMS: A dropdown menu with '-Select-'.
- BUDGET RANGE: A dropdown menu with '-Select-'.
- PROPERTY VALUE: A text input field with a placeholder '##,##,###,###'.

On the right side, there is a 'Form Customization - Web' panel with 'Label placement' options. The bottom of the interface shows a status bar with 'Trial has expired | Upgrade' and a 'Help' button.

The screenshot displays the Zoho Creator form builder interface. On the left, there's a 'Basic Fields' panel with icons for Name, Email, Address, Phone, Single Line, Multi Line, Number, Date, Time, and Drop Down. The central workspace contains several field cards: TYPE (selected), PROPERTY TYPE, LOCALITY, AREA(sq.ft), BHK/ROOMS, BUDGET RANGE, PROPERTY VALUE, RENTAL AGREEMENT, DURATION, and FURNISHING STATUS. On the right, the 'Field Properties' panel for the 'TYPE' field is visible, showing options for 'BUY', 'RENT', and 'COMMERCIAL' under the 'Choices' section. The interface also includes a top navigation bar with 'Home - Classroom', 'Zoho Creator', and 'Ask Gemini' buttons, and a bottom status bar indicating 'Trial has expired'.

The supplied implementation evidence shows the Property Buying and Selling application with a form containing fields including TYPE, PROPERTY TYPE, LOCALITY, AREA (sq.ft), BHK/ROOMS, BUDGET RANGE, PROPERTY VALUE, RENTAL AGREEMENT, DURATION and FURNISHING STATUS. The TYPE field is configured with choices including BUY and RENT, demonstrating the central transaction selection required by the project.

## 12. FORMS AND DATA ENTRY

| Form                  | Important Fields / Purpose                                                                                                |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------|
| Property Registration | Property ID, listing type, category, property type, title, locality, area, BHK/rooms, value/rent, availability and status |
| Buyer                 | Name, email/phone, preferred location, budget and property preference                                                     |
| Tenant                | Name, contact, preferred locality, rent budget and rental preference                                                      |
| Property Enquiry      | Property ID, user, enquiry type, message and enquiry status                                                               |
| Visit Request         | Property ID, requester, preferred date/time and visit status                                                              |
| Rental Agreement      | Agreement ID, property, owner, tenant, rent, advance, dates, terms and renewal                                            |



## 12.1 Main Property Form Evidence

The screenshot shows a web application interface for 'PROPERTY BUYING AND RENTAL'. The sidebar on the left contains the following items:

- PROPERTY BUYING AND RENTAL
- PROPERTY BUYING AND RENTAL
- PROPERTY BUYING AND RENTAL Report
- DASHBOARD

The main content area displays a table titled 'PROPERTY BUYING AND RENTAL Report'. The table has the following columns:

| TYPE | PROPERTY TYPE     | LOCALITY    | AREA(sq.ft) | BHK/ROOMS | BUDGET RANGE | PROPERTY VALUE   |
|------|-------------------|-------------|-------------|-----------|--------------|------------------|
| RENT | WARE HOUSE        | TAMBARAM    | 1036        | 3BHK      | 50-75L       | ₹ 1,50,00,000.00 |
| RENT | WARE HOUSE        | KILAMBAKKAM | 3023        | 4BHK      | 1Cr+         | ₹ 1,50,00,000.00 |
| RENT | INDEPENDENT HOUSE | ADAYAR      | 4567        | 3BHK      | 75L-1Cr      | ₹ 1,50,00,000.00 |

The interface also includes a search bar with the filter 'TYPE contains RENT', buttons for 'Save Changes' and 'Remove Changes', and a status bar at the bottom indicating 'Showing 3 of 3'.

## 13. VALIDATION AND DATA QUALITY RULES

| Validation        | Rule / Expected Behaviour                                                 |
|-------------------|---------------------------------------------------------------------------|
| Property ID       | Should be unique and non-empty.                                           |
| Listing Type      | Should be BUY or RENT.                                                    |
| Property Category | Should be RESIDENTIAL or COMMERCIAL.                                      |
| Area              | Should be positive.                                                       |
| BHK/Rooms         | Should be a valid non-negative value where applicable.                    |
| Property Value    | Required/positive for sale listings.                                      |
| Monthly Rent      | Required/positive for rental listings.                                    |
| Advance/Deposit   | Should not be negative.                                                   |
| Agreement Dates   | End date should not precede start date.                                   |
| Property Status   | Should use controlled choices such as Available, Booked, Sold and Rented. |
| Contact           | Should follow the expected contact format.                                |
| Locality          | Should use consistent Chennai locality names for reliable analytics.      |

Validation reduces incomplete, contradictory and duplicate information. It is particularly important for Big Data Analytics because inconsistent locality names or missing values can distort counts and averages.

## 14. WORKFLOW, AUTOMATION AND COLLABORATION

The workflow layer connects data entry with application behaviour. The objective is to reduce repetitive manual updates and keep the property status consistent with the transaction process.

| Trigger / Event   | Workflow Action                                                   |
|-------------------|-------------------------------------------------------------------|
| Listing created   | Set initial property status to Available where appropriate.       |
| Buy selected      | Display/validate sale-related fields such as property value.      |
| Rent selected     | Display/validate rent, advance and agreement-related information. |
| Enquiry submitted | Create enquiry record and set enquiry status.                     |
| Visit requested   | Record visit details and maintain visit status.                   |
| Property booked   | Update property status to Booked.                                 |
| Sale completed    | Update property status to Sold.                                   |
| Rental completed  | Update property status to Rented and associate agreement.         |
| Listing updated   | Refresh report information available to authorized users.         |

The screenshot shows the Zoho Creator workflow builder interface for a form titled 'PROPERTY BUYING AND RENTAL'. The breadcrumb navigation indicates the path: 'PROPERTY BUYING AND RENTAL > Created > Field rules > RENTAL'. A note states: 'Actions will be executed on form load and on user input of any of the fields participating in the condition. [Learn more](#)'. The workflow consists of two conditional actions:

- Action 1:** 'Execute only if TYPE\_field == "RENT"'. The action is 'Show fields [BHK, ROOMS, FURNISHING\_STATUS, RENTAL\_AGREEMENT]'.
- Action 2:** 'Execute only if RENTAL\_AGREEMENT == "OTHER"'. The action is 'Show fields [DURATION]'.

At the bottom, there is an 'Add New Action' button. The footer of the interface includes 'My Pins', 'Chats', 'Contacts', a 'Trial has expired | Upgrade' message, and a 'Help' link.



| TYPE       | PROPERTY TYPE     | LOCALITY    | AREA(sq.ft) | BHK/ROOMS | BUDGET RANGE | PROPERTY VALUE   |
|------------|-------------------|-------------|-------------|-----------|--------------|------------------|
| RENT       | WARE HOUSE        | TAMBARAM    | 1036        | 3BHK      | 50-75L       | ₹ 1,50,00,000.00 |
| RENT       | WARE HOUSE        | KILAMBAKKAM | 3023        | 4BHK      | 1Cr+         | ₹ 1,50,00,000.00 |
| RENT       | INDEPENDENT HOUSE | ADAYAR      | 4567        | 3BHK      | 75L-1Cr      | ₹ 1,50,00,000.00 |
| COMMERCIAL | APARTMENT         | VELACHERY   | 6253        |           | 50-75L       | ₹ 1,50,00,000.00 |
| BUY        | APARTMENT         | VELACHERY   |             |           | UNDER 25L    | ₹ 1,50,00,000.00 |
| COMMERCIAL | INDEPENDENT HOUSE | OMR         | 625651      |           | 25-50L       | ₹ 1,50,00,000.00 |
| BUY        | VILLA             | OMR         | 12000       |           | 1Cr+         | ₹ 1,50,00,000.00 |

## 14.1 Cloud Collaboration

Because the application is hosted as a cloud service, authorized users can work with the same centralized application data. Owners/agents can update listings, while buyers/tenants can access current information through the application interface. Role-based access is important so that users only perform operations appropriate to their responsibilities.

## 15. SEARCH, REPORTS AND DASHBOARD

Reports convert stored property records into useful views. The application can organize listings by transaction type, category, locality, budget and rental agreement information.

| Report / View          | Purpose                                           |
|------------------------|---------------------------------------------------|
| Sale Properties        | Shows properties available for buying.            |
| Rental Properties      | Shows properties available for rent.              |
| Residential Properties | Filters residential listings.                     |
| Commercial Properties  | Filters commercial listings.                      |
| Area-wise Listings     | Groups or filters properties by Chennai locality. |

|                      |                                                            |
|----------------------|------------------------------------------------------------|
| Budget-wise Listings | Supports comparison based on budget range.                 |
| Rental Agreements    | Shows agreement records and rental details.                |
| Property Status      | Shows Available, Booked, Sold and Rented properties.       |
| Dashboard            | Combines important summary views for quick interpretation. |

The screenshot displays a web application interface for 'PROPERTY BUYING AND RENTAL'. The main content area shows a table titled 'PROPERTY BUYING AND RENTAL Report' with the following data:

| TYPE       | PROPERTY TYPE     | LOCALITY    | AREA(sq.ft) | BHK/ROOMS | BUDGET RANGE | PROPERTY VALUE   |
|------------|-------------------|-------------|-------------|-----------|--------------|------------------|
| RENT       | WARE HOUSE        | TAMBARAM    | 1036        | 3BHK      | 50-75L       | ₹ 1,50,00,000.00 |
| RENT       | WARE HOUSE        | KILAMBAKKAM | 3023        | 4BHK      | 1Cr+         | ₹ 1,50,00,000.00 |
| RENT       | INDEPENDENT HOUSE | ADAYAR      | 4567        | 3BHK      | 75L-1Cr      | ₹ 1,50,00,000.00 |
| COMMERCIAL | APARTMENT         | VELACHERY   | 6253        |           | 50-75L       | ₹ 1,50,00,000.00 |
| BUY        | APARTMENT         | VELACHERY   |             |           | UNDER 25L    | ₹ 1,50,00,000.00 |
| COMMERCIAL | INDEPENDENT HOUSE | OMR         | 625651      |           | 25-50L       | ₹ 1,50,00,000.00 |
| BUY        | VILLA             | OMR         | 12000       |           | 1Cr+         | ₹ 1,50,00,000.00 |

The interface includes a sidebar with navigation options: PROPERTY BUYING AND RENTAL, PROPERTY BUYING AND RENTAL Report, and DASHBOARD. The bottom status bar shows 'Showing 7 of 7'.

The uploaded implementation evidence identifies both a Report and Dashboard component for the Property Buying and Selling work.

## 16. CHENNAI SAMPLE PROPERTY DATASET

A sample dataset is required to demonstrate Big Data preparation and MapReduce analytics. The dataset should contain multiple Chennai localities and a mixture of sale and rental listings. The following format is suitable for the project.

| Property ID | Type | Category    | Property Type | Locality   | Area sq.ft | BHK | Value / Rent | Status    |
|-------------|------|-------------|---------------|------------|------------|-----|--------------|-----------|
| CHN001      | BUY  | RESIDENTIAL | Flat          | Anna Nagar | 1200       | 2   | ₹8500000     | Available |
| CHN002      | RENT | RESIDENTIAL | House         | Ambattur   | 1100       | 2   | ₹22000       | Available |
| CHN003      | BUY  | RESIDENTIAL | Villa         | Porur      | 2100       | 3   | ₹12500000    | Booked    |
| CHN004      | RENT | COMMERCIAL  | Office        | OMR        | 1800       | -   | ₹65000       | Available |
| CHN005      | BUY  | RESIDENTIAL | Flat          | Velachery  | 950        | 2   | ₹7200000     | Available |
| CHN006      | RENT | RESIDENTIAL | Flat          | Tambaram   | 900        | 2   | ₹18000       | Rented    |
| CHN007      | BUY  | COMMERCIAL  | Shop          | T Nagar    | 700        | -   | ₹15000000    | Available |
| CHN008      | RENT | RESIDENTIAL | Flat          | Adyar      | 1250       | 3   | ₹42000       | Available |

|        |      |             |        |       |      |   |          |           |
|--------|------|-------------|--------|-------|------|---|----------|-----------|
| CHN009 | BUY  | RESIDENTIAL | Flat   | Porur | 1050 | 2 | ₹6800000 | Available |
| CHN010 | RENT | COMMERCIAL  | Office | OMR   | 2500 | - | ₹90000   | Available |

The table is a small illustrative sample. For the final analytics demonstration, additional rows should be prepared so that repeated localities, property types and transaction types exist and meaningful aggregation can be observed.

## 17. BIG DATA PROBLEM ANALYSIS

| Big Data Dimension / Issue | Property Domain Example                                                  | Management Approach                                                  |
|----------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------|
| Volume                     | Large numbers of listings, searches, enquiries and transactions.         | Scalable storage and partitioned/aggregated analytics.               |
| Velocity                   | Frequent listing, status, enquiry and search updates.                    | Workflow automation and timely processing.                           |
| Variety                    | Structured property fields, text descriptions, location data and images. | Structured schema plus appropriate handling of different data types. |
| Veracity                   | Duplicate, incomplete, outdated or incorrectly entered listings.         | Validation, standardization and duplicate detection.                 |
| Location Data              | Multiple locality/address representations.                               | Standardize Chennai locality names and preserve location fields.     |
| Privacy                    | Owner, tenant and contact information.                                   | Role-based access and controlled exposure of personal data.          |
| Scalability                | Dataset may grow as more properties and users are added.                 | Cloud storage and distributed processing concepts.                   |
| Duplicates                 | Same property may be listed multiple times.                              | Property ID, owner checks and duplicate-detection rules.             |

The academic sample dataset is not itself industrial-scale Big Data. However, the project models the data-management and analytics issues that become significant when the property platform grows to large numbers of listings, user interactions and historical records.

## 18. MAPREDUCE APPLICATION DESIGN

MapReduce is suitable for aggregation-style property analytics because the mapper can convert each property record into a key-value pair and the reducer can combine values belonging to the same key.

### 18.1 Analytics Task A – Area-wise Listing Count

Mapper: read each property record and emit (Locality, 1). Reducer: sum all values for each locality to obtain the number of listings.

### 18.2 Analytics Task B – Property-type Demand

Mapper: emit (Property Type, 1) for each relevant listing or search record. Reducer: sum the counts to identify the most represented/demanded property type in the sample.

## 18.3 Analytics Task C – Average Rental Value

Mapper: for rental records emit (Locality, (Rent, 1)). Reducer: calculate total rent and number of records for each locality, then compute average rent as total rent divided by count.

| Metric                  | Mapper Key    | Mapper Value | Reducer Output     |
|-------------------------|---------------|--------------|--------------------|
| Area-wise listing count | Locality      | 1            | Total listings     |
| Property-type demand    | Property Type | 1            | Total demand/count |
| Average rent            | Locality      | (Rent,1)     | Average rent       |
| Average sale value      | Locality      | (Value,1)    | Average sale value |

## 19. MAPREDUCE PSEUDOCODE AND SOURCE CODE

### 19.1 Pseudocode – Area-wise Count

```
MAP(record):  
    locality = record.Locality  
    EMIT(locality, 1)
```

```
REDUCE(locality, values):  
    total = SUM(values)  
    EMIT(locality, total)
```

### 19.2 Python-Style Demonstration

```
from collections import defaultdict
```

```
def mapper(records):  
    for record in records:  
        yield record["Locality"], 1
```

```
def reducer(mapped):  
    totals = defaultdict(int)  
    for locality, value in mapped:  
        totals[locality] += value  
    return dict(totals)
```

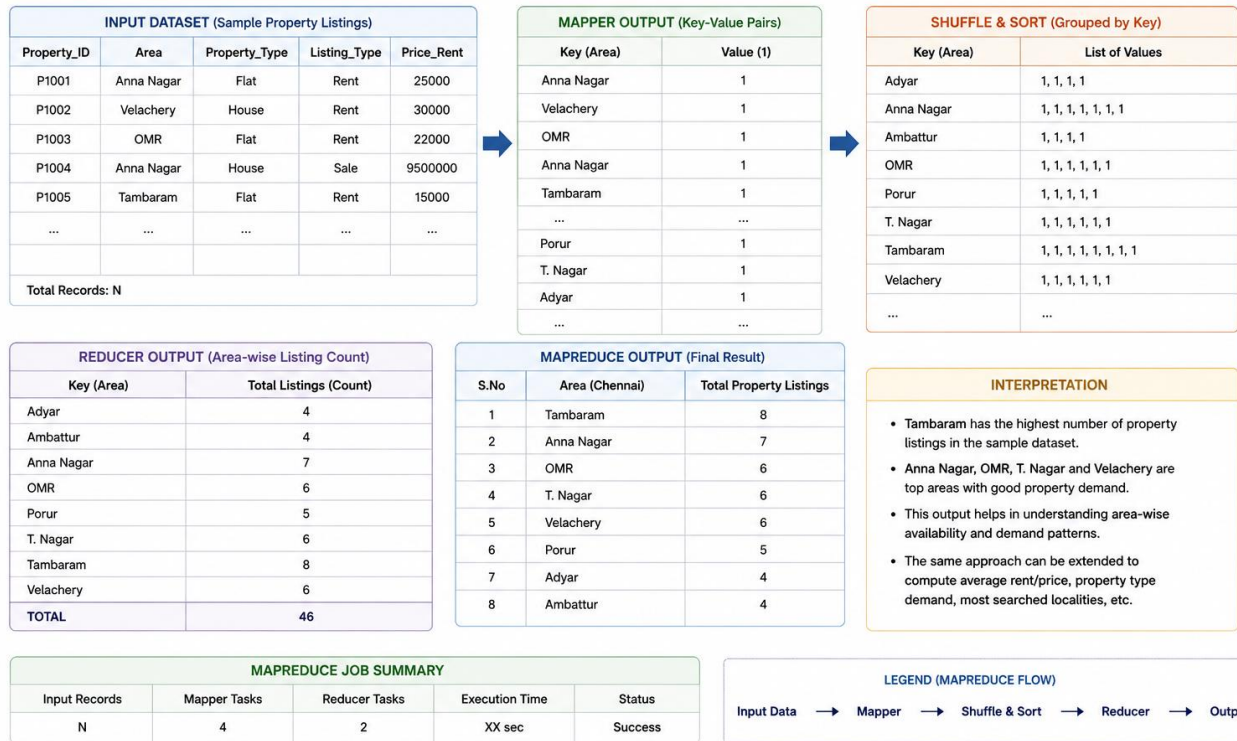
### 19.3 Average Rent Logic

```
MAP(record):  
    if record.ListingType == "RENT":  
        EMIT(record.Locality, (record.MonthlyRent, 1))
```

```
REDUCE(locality, pairs):  
    total_rent = SUM(pair.rent)  
    count = SUM(pair.count)  
    average = total_rent / count  
    EMIT(locality, average)
```

## FIGURE 19.2 – MAPREDUCE OUTPUT

Analytics: Area-wise Property Listing Count in Chennai



## 20. ALGORITHM AND FLOWCHART

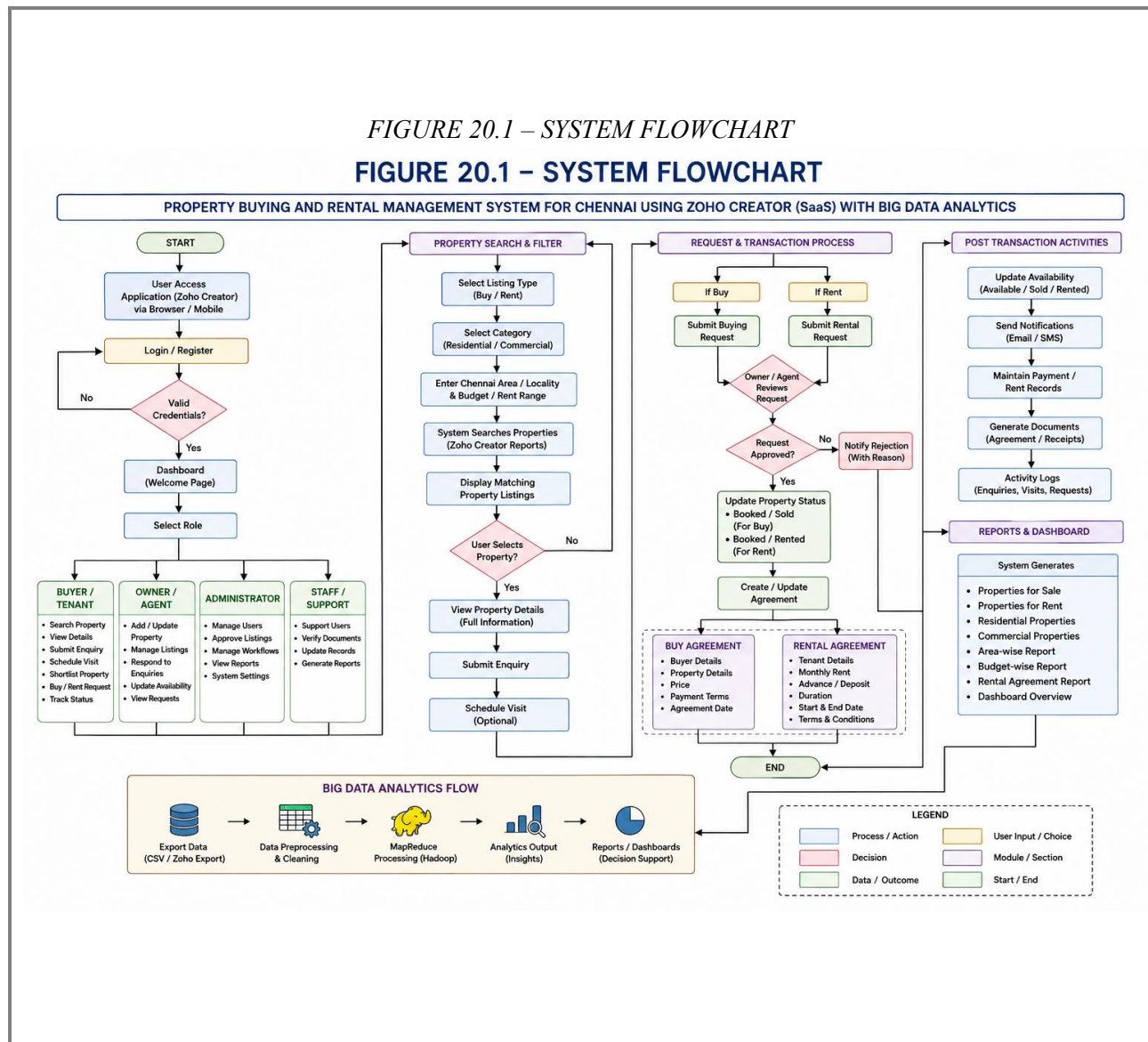
### 20.1 Property Listing Algorithm

21. Start.
22. Enter property details.
23. Validate Property ID and required fields.
24. Select BUY or RENT.
25. Select RESIDENTIAL or COMMERCIAL.
26. Enter locality and property details.
27. If RENT, validate rent, advance and agreement-related fields.
28. Set initial property status.
29. Save the property record.
30. Display the property in the appropriate report.
31. End.

## 20.2 Analytics Algorithm

32. Load the prepared Chennai property dataset.
33. Clean missing, duplicate and inconsistent records.
34. Select the required analytics metric.
35. Mapper emits key-value pairs.
36. Group records by key.
37. Reducer aggregates values.
38. Generate analytical output.
39. Interpret locality/property-type/price-rent patterns.
40. End.

FIGURE 20.1 – SYSTEM FLOWCHART





## 21. TESTING AND VALIDATION

Testing verifies that the application accepts valid information, rejects invalid values and produces expected reports and analytical results.

| TC ID | Test Case               | Input / Condition          | Expected Result                           | Actual Result                |
|-------|-------------------------|----------------------------|-------------------------------------------|------------------------------|
| TC01  | Create valid property   | All mandatory fields valid | Record saved                              | To be filled after execution |
| TC02  | Invalid area            | Area = negative            | Validation/error                          | To be filled after execution |
| TC03  | Buy listing             | Listing Type = BUY         | Sale fields available/validated           | To be filled after execution |
| TC04  | Rent listing            | Listing Type = RENT        | Rent/agreement fields available/validated | To be filled after execution |
| TC05  | Locality search         | Locality = OMR             | Matching listings displayed               | To be filled after execution |
| TC06  | Status update           | Property marked Sold       | Status becomes Sold                       | To be filled after execution |
| TC07  | Rental agreement        | Valid agreement dates      | Agreement record accepted                 | To be filled after execution |
| TC08  | Invalid agreement dates | End date before start date | Validation/error                          | To be filled after execution |
| TC09  | Dashboard               | Valid records present      | Dashboard/report values displayed         | To be filled after execution |
| TC10  | MapReduce               | Prepared dataset loaded    | Expected aggregation generated            | To be filled after execution |

PROPERTY BUYING AND RENTAL Report

| TYPE       | PROPERTY TYPE     | LOCALITY    | AREA(sq.ft) | BHK/ROOMS | BUDGET RANGE | PROPERTY VALUE   |
|------------|-------------------|-------------|-------------|-----------|--------------|------------------|
| RENT       | WARE HOUSE        | TAMBARAM    | 1036        | 3BHK      | 50-75L       | ₹ 1,50,00,000.00 |
| RENT       | WARE HOUSE        | KILAMBAKKAM | 3023        | 4BHK      | 1Cr+         | ₹ 1,50,00,000.00 |
| RENT       | INDEPENDENT HOUSE | ADAYAR      | 4567        | 3BHK      | 75L-1Cr      | ₹ 1,50,00,000.00 |
| COMMERCIAL | APARTMENT         | VELACHERY   | 6253        |           | 50-75L       | ₹ 1,50,00,000.00 |
| BUY        | APARTMENT         | VELACHERY   |             |           | UNDER 25L    | ₹ 1,50,00,000.00 |
| COMMERCIAL | INDEPENDENT HOUSE | OMR         | 625651      |           | 25-50L       | ₹ 1,50,00,000.00 |
| BUY        | VILLA             | OMR         | 12000       |           | 1Cr+         | ₹ 1,50,00,000.00 |

Showing 7 of 7

## 22. RESULTS AND DISCUSSION

The implementation demonstrates the core structure of a cloud-based property management application. The Zoho Creator evidence supplied for the project shows the Property Buying and Selling application and identifies report and dashboard components. The property form includes transaction-related choices and several property information fields, establishing the base for a centralized listing system.

The expected functional result is that property information can be stored in structured records and viewed through reports. Buy/Rent classification supports separate sale and rental handling, while property type, locality, area, BHK/rooms, budget and furnishing fields support filtering and comparison.

The Big Data component extends the system beyond simple record storage. Repeated locality and property-type records can be aggregated to identify which Chennai areas or property categories have higher listing counts. Rental records can also be summarized to estimate average rent by locality.

Actual execution values, screenshots and measured analytics should be inserted after the final test run. The report intentionally leaves these evidence areas open rather than claiming unverified numerical performance.

## 23. ANALYSIS, COMPARISON, TRADE-OFFS AND JUSTIFICATION

| Approach                      | Advantages                                                                               | Trade-offs                                                                                          |
|-------------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Manual/spreadsheet management | Simple for very small datasets.                                                          | Harder collaboration, duplicate records, limited workflow and weaker scalability.                   |
| Custom web application        | High control over UI, backend and algorithms.                                            | Requires more development, hosting, security and maintenance effort.                                |
| Zoho Creator SaaS             | Fast cloud development, forms, workflows, reports and reduced infrastructure management. | Platform capabilities and customization depend on service features/plan.                            |
| Centralized property dataset  | Consistent source for reports and analytics.                                             | Requires careful validation, privacy and duplicate management.                                      |
| MapReduce aggregation         | Simple scalable model for key-based aggregation.                                         | Not necessary for tiny datasets; setup/processing overhead is higher than direct local computation. |

Zoho Creator is justified for this assignment because the focus is on demonstrating SaaS, cloud access, structured forms, workflows, reporting and analytics without requiring the student to build the entire infrastructure stack. MapReduce is justified as a course-aligned demonstration of distributed aggregation rather than because the sample dataset itself requires distributed processing.

## 24. ADVANTAGES

- Centralized property information.
- Cloud-based access through the SaaS application.
- Structured Buy/Rent and Residential/Commercial classification.
- Chennai locality-based organization.
- Improved data consistency through validation.
- Workflow support for property status and enquiries.
- Reports and dashboards for quick information access.
- Rental Agreement information can be linked to properties.
- Big Data analysis can identify useful patterns from growing records.
- MapReduce demonstrates scalable aggregation concepts.
- Reduced infrastructure-management effort compared with building a complete application stack.

## 25. LIMITATIONS AND POSSIBLE IMPROVEMENTS

### 25.1 Limitations

- The classroom dataset is smaller than a real industrial property dataset.
- External real-time property feeds are not included.
- Advanced map/GIS integration is outside the core prototype.
- Image-based property similarity or recommendation is not implemented in the core system.
- Real-world payment, legal verification and identity verification are outside scope.
- Analytics accuracy depends on the completeness and consistency of the dataset.
- Some advanced Zoho features may depend on account capabilities.

### 25.2 Possible Improvements

- Integrate maps and geospatial search for Chennai localities.
- Add duplicate-listing detection using Property ID, address and owner information.
- Add recommendation models based on user preferences.
- Introduce stronger role-based access and audit logs.
- Add notification/email automation for enquiries and agreement events.
- Expand the dataset with historical searches, visits and transactions.
- Use distributed storage and processing for genuinely large datasets.
- Add visual analytics for locality demand, rent trends and property-type demand.
- Integrate document upload and verification for rental agreements.
- Add secure authentication and privacy controls for production deployment.

## 26. BROADER CONSIDERATIONS AND SDG RELEVANCE

| SDG                                             | Relevance to Project                                                                                              |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| SDG 9 – Industry, Innovation and Infrastructure | Demonstrates cloud-based digital infrastructure and technology-enabled property management.                       |
| SDG 11 – Sustainable Cities and Communities     | Organized property information can support better access to housing and urban property resources.                 |
| SDG 12 – Responsible Consumption and Production | Digital records and analytics can reduce repetitive manual handling and support more informed property decisions. |

The SDG connection is primarily conceptual. The project demonstrates how digital infrastructure and data analytics can support urban information management; it does not claim direct measurement of SDG indicators.

## 27. CONCLUSION

The Property Buying and Rental Management System for Chennai demonstrates the design of a cloud-based SaaS solution using Zoho Creator. The system organizes property listings and related information through structured forms, supports Buy/Rent and Residential/Commercial classification, and provides a foundation for enquiries, visits, rental agreements, property status management, reports and dashboards.

The project also connects cloud application development with Big Data Analytics. A prepared Chennai property dataset can be analysed for locality and property-type patterns, while MapReduce provides a course-aligned approach for aggregation. The project highlights the importance of data quality, duplicate handling, privacy, scalability and location consistency when property information grows.

Overall, the work provides practical exposure to cloud service models, virtualization concepts, cloud access and collaboration, Big Data challenges and MapReduce-based analytics. Further development can add richer geospatial services, recommendations, stronger security and larger-scale distributed analytics.

## 28. CO MAPPING AND INDIVIDUAL CONTRIBUTION

| Course Outcome | Evidence / Contribution                                                                                                         |
|----------------|---------------------------------------------------------------------------------------------------------------------------------|
| CO1            | Explained IaaS, PaaS and SaaS; justified Zoho Creator as SaaS; implemented structured property forms.                           |
| CO2            | Designed conceptual virtualization and cloud infrastructure architecture supporting the SaaS application.                       |
| CO3            | Applied cloud-based access, shared data, workflows, reports, dashboard and collaboration concepts.                              |
| CO4            | Analysed property Big Data issues including volume, velocity, variety, veracity, duplicates, location, privacy and scalability. |
| CO5            | Designed MapReduce mapper/reducer logic for area-wise counts, property-type demand and price/rent analytics.                    |

## 28.1 Individual Contribution

| Student                   | Contribution                                                                                                                                                                                                                                                                            |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Logeshwaran R (192511175) | Problem analysis, system design, Zoho Creator application configuration, property data model, form planning, validation/workflow design, report/dashboard planning, sample dataset preparation, Big Data analysis, MapReduce methodology, testing documentation and report preparation. |

## 29. INDIVIDUAL REFLECTION

Working on this project helped me understand how cloud computing concepts can be applied to a practical application rather than being treated only as theoretical topics. Designing the property management system required converting a real-world problem into structured entities, fields, relationships and workflows. The Buy/Rent and Residential/Commercial selections showed how application logic can be organized around user requirements.

Using Zoho Creator provided practical exposure to SaaS because the application could be configured through cloud-based tools without requiring complete infrastructure management. I learned that good data modelling is important before creating forms because field choices, relationships and validation rules affect the quality of reports and analytics.

The Big Data portion helped me understand that large datasets create challenges beyond storage. Duplicate records, inconsistent locality names, missing values, privacy and rapid updates can all affect analytical accuracy. The MapReduce exercise showed how a large collection of records can be transformed into key-value pairs and aggregated into useful summaries.

One of the important challenges was connecting the application part with the analytics part. The solution is to maintain a clean and consistent dataset so that records exported or prepared for analysis can be processed reliably. This project also improved my understanding of how cloud applications and data analytics can contribute to urban services, especially in the context of SDG 9, SDG 11 and SDG 12.

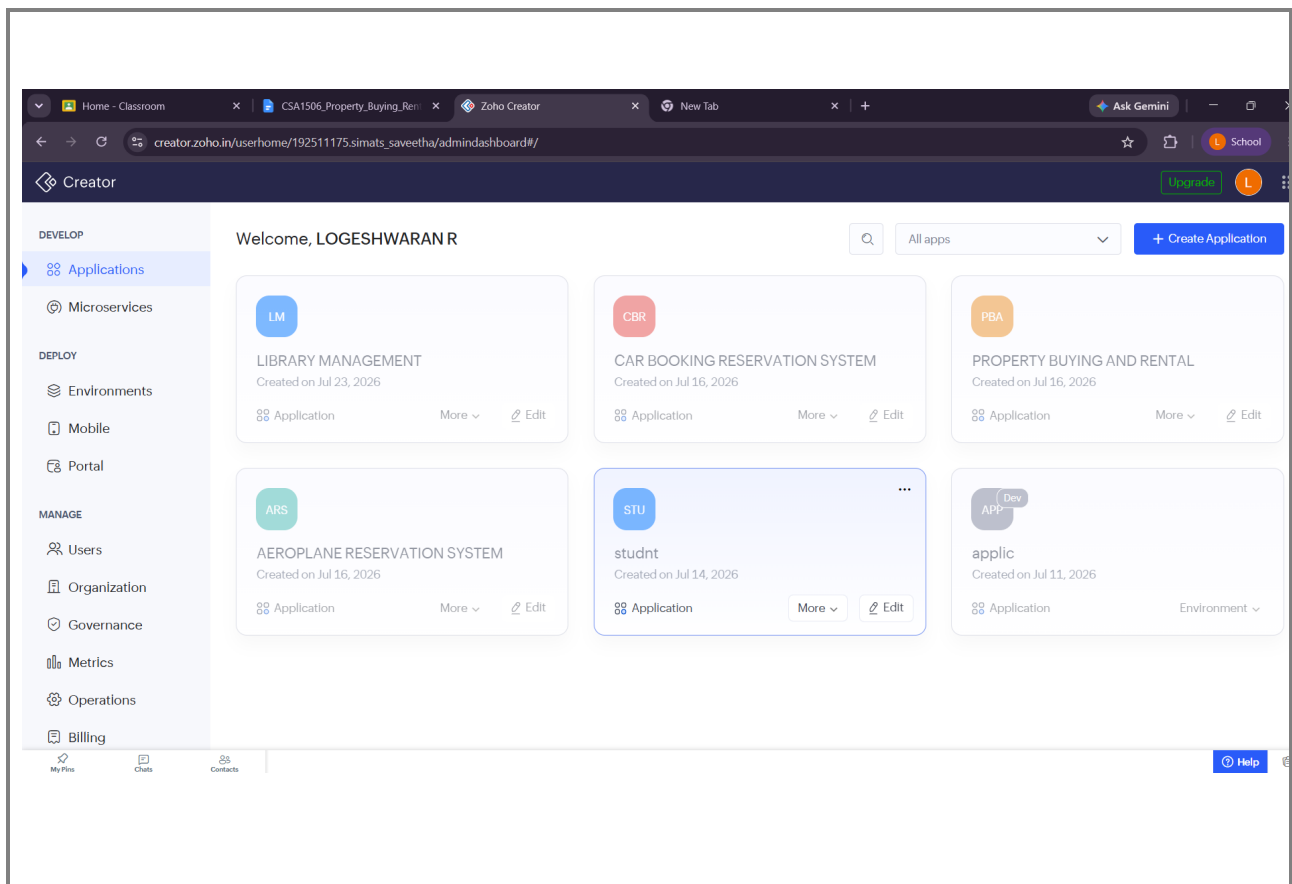
CO1 was achieved through understanding cloud service models and applying SaaS. CO2 was addressed through virtualization and infrastructure design. CO3 was addressed through cloud access, collaboration and reporting. CO4 was addressed through Big Data problem analysis. CO5 was addressed through MapReduce design and property analytics.

## 30. REFERENCES

1. National Institute of Standards and Technology (NIST), The NIST Definition of Cloud Computing, Special Publication 800-145.
2. Zoho Creator documentation and product learning resources, used for understanding application forms, workflows, reports and dashboards.
3. J. Dean and S. Ghemawat, "MapReduce: Simplified Data Processing on Large Clusters," Communications of the ACM, 2008.

4. Apache Hadoop documentation, MapReduce concepts and programming model.
5. United Nations, Sustainable Development Goals – SDG 9, SDG 11 and SDG 12.
6. CSA1506 – Cloud Computing and Big Data Analytics assignment instructions and course outcomes.
7. Student-developed Zoho Creator application: Property Buying and Selling / Property Buying and Rental Management System.
8. Student-prepared Chennai property sample dataset for academic analytics demonstration.

## APPENDIX A – SCREENSHOT PLACEHOLDERS



Home - Classroom

CSA1506\_Property\_Buying\_Ren

Zoho Creator

New Tab

Ask Gemini

creatorapp.zoho.in/appbuilder/192511175.simats\_saveetha/property-buying-and-rental/formbuilder/PROPERTY\_BUYING\_AND\_RENTAL/edit

PROPERTY BUYING AND RENTAL

PROPERTY BUYING AND R...

Done

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

123

Number

Date

Time

Drop Down

TYPE

PROPERTY TYPE

LOCALITY

AREA(sq.ft)

BHK/ROOMS

BUDGET RANGE

PROPERTY VALUE

RENTAL AGREEMENT

DURATION

FURNISHING STATUS

Field Properties

Field name

TYPE

Field link name

TYPE\_field

View Field References

Choices

Advanced

Renaming the choices will update the values in the existing records, if any.

BUY

RENT

COMMERCIAL

Alphabetical Order

Allow Other Choice

My Pins

Clubs

Contacts

Trial has expired | Upgrade

Help

Home - Classroom

CSA1506\_Property\_Buying\_Ren

192511175.simats\_saveetha -

New Tab

Ask Gemini

creatorapp.zoho.in/192511175.simats\_saveetha/property-buying-and-rental/#Report:PROPERTY\_BUYING\_AND\_RENTAL\_Report

PROPERTY BUYING AND RENTAL

Trial has expired Upgrade

Edit this application

Help

PROPERTY BUYING AND RENTAL

PROPERTY BUYING AND RENTAL

PROPERTY BUYING AND RENTAL

PROPERTY BUYING AND RENTAL Report

DASHBOARD

LOGESHWA...

PROPERTY BUYING AND RENTAL Report

Save Changes

Remove Changes

Search

TYPE contains either "BUY" or "RENT"

| TYPE | PROPERTY TYPE     | LOCALITY    | AREA(sq.ft) | BHK/ROOMS | BUDGET RANGE | PROPERTY VALUE   |
|------|-------------------|-------------|-------------|-----------|--------------|------------------|
| RENT | WARE HOUSE        | TAMBARAM    | 1036        | 3BHK      | 50-75L       | ₹ 1,50,00,000.00 |
| RENT | WARE HOUSE        | KILAMBAKKAM | 3023        | 4BHK      | 1Cr+         | ₹ 1,50,00,000.00 |
| RENT | INDEPENDENT HOUSE | ADAYAR      | 4567        | 3BHK      | 75L-1Cr      | ₹ 1,50,00,000.00 |
| BUY  | APARTMENT         | VELACHERY   |             |           | UNDER 25L    | ₹ 1,50,00,000.00 |
| BUY  | VILLA             | OMR         | 12000       |           | 1Cr+         | ₹ 1,50,00,000.00 |

Showing 5 of 5

#### A4. PROPERTY DATA ENTRY

The screenshot displays a web browser window with the URL `creatorapp.zoho.in/192511175.simats_saveetha/property-buying-and-rental/#Report:PROPERTY_BUYING_AND_RENTAL_Report`. The page title is "PROPERTY BUYING AND RENTAL".

**Mobile App Preview (Left):** A smartphone screen shows the app interface with the following fields:

- TYPE:** Radio buttons for Choice 1 (selected), Choice 2, and Choice 3.
- PROPERTY TYPE:** A dropdown menu with "Select value".
- LOCALITY:** A dropdown menu with "Select value".
- AREA(sq.ft):** A text input field with "Enter value".
- BHK/ROOMS:** A dropdown menu with "Select value".
- BUDGET RANGE:** A dropdown menu with "Select value".

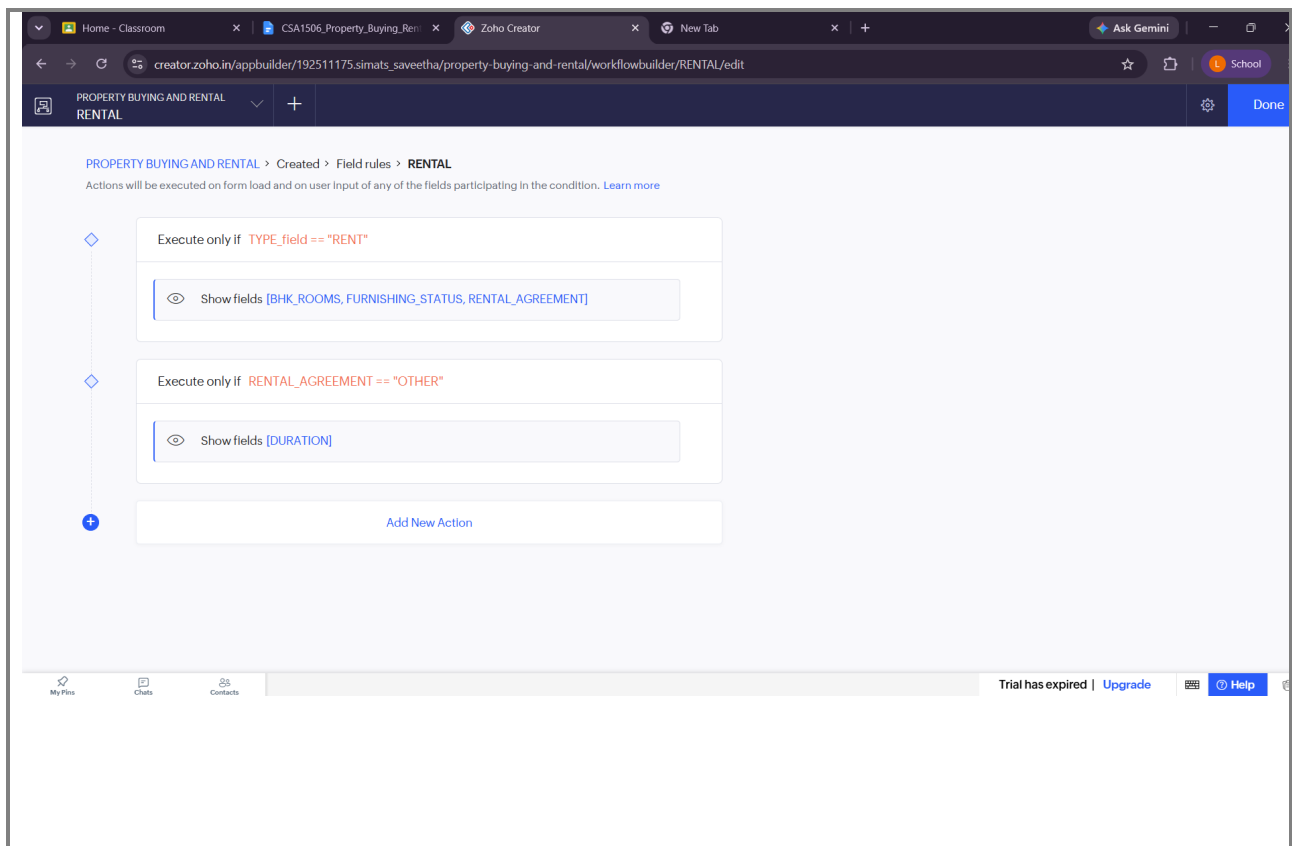
**Zoho Creator on Mobile (Right):** A promotional banner with the text "Zoho Creator on Mobile" and "The Zoho Creator mobile app lets you use your web apps on mobile devices." It lists the following benefits:

- Optimize your apps with mobile gestures
- Use your device's camera to scan bar codes and QR codes
- Geofence forms and geotag where records are submitted
- Access, update, and analyze data faster

Below the list are buttons for "Download on the App Store" and "GET IT ON Google Play". At the bottom, there is an input field for "Enter email address" and a "Send Link" button.

#### A10. VALIDATION RULE / WORKFLOW





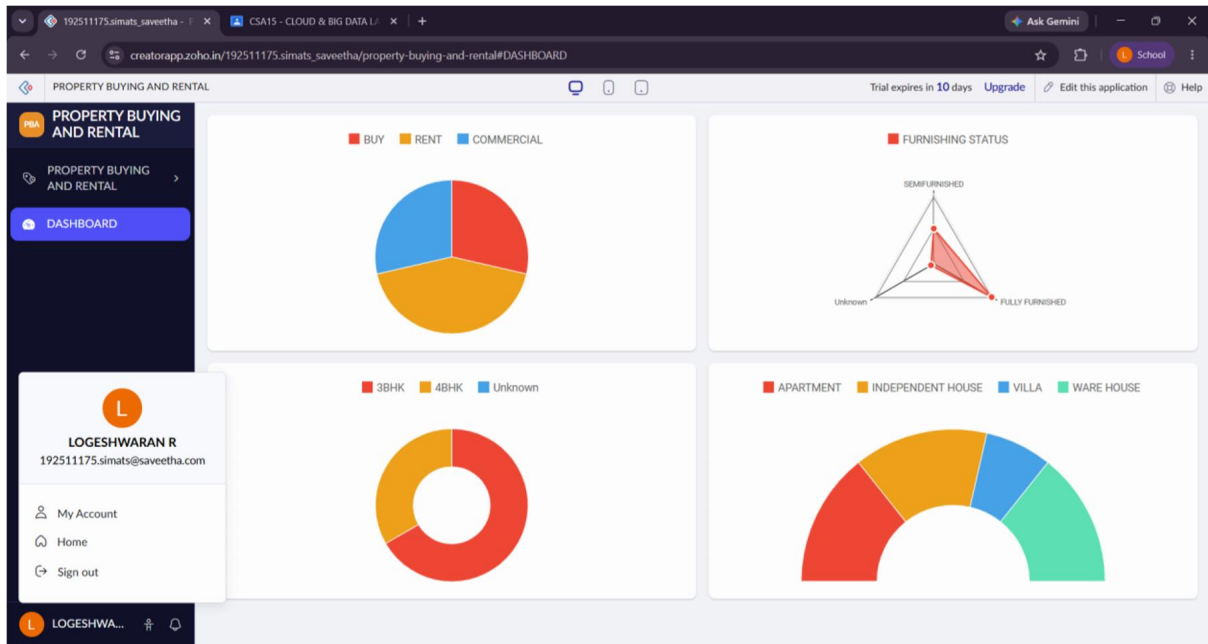
## 111. PROPERTY REPORT

The screenshot displays the 'PROPERTY BUYING AND RENTAL Report' in Zoho Creator. The report title is 'PROPERTY BUYING AND RENTAL Report'. The table below lists property details with columns for TYPE, PROPERTY TYPE, LOCALITY, AREA(sq.ft), BHK/ROOMS, BUDGET RANGE, and PROPERTY VALUE.

| TYPE       | PROPERTY TYPE     | LOCALITY    | AREA(sq.ft) | BHK/ROOMS | BUDGET RANGE | PROPERTY VALUE   |
|------------|-------------------|-------------|-------------|-----------|--------------|------------------|
| RENT       | WARE HOUSE        | TAMBARAM    | 1036        | 3BHK      | 50-75L       | ₹ 1,50,00,000.00 |
| RENT       | WARE HOUSE        | KILAMBAKKAM | 3023        | 4BHK      | 1Cr+         | ₹ 1,50,00,000.00 |
| RENT       | INDEPENDENT HOUSE | ADAYAR      | 4567        | 3BHK      | 75L-1Cr      | ₹ 1,50,00,000.00 |
| COMMERCIAL | APARTMENT         | VELACHERY   | 6253        |           | 50-75L       | ₹ 1,50,00,000.00 |
| BUY        | APARTMENT         | VELACHERY   |             |           | UNDER 25L    | ₹ 1,50,00,000.00 |
| COMMERCIAL | INDEPENDENT HOUSE | OMR         | 625651      |           | 25-50L       | ₹ 1,50,00,000.00 |
| BUY        | VILLA             | OMR         | 12000       |           | 1Cr+         | ₹ 1,50,00,000.00 |

The left sidebar shows navigation options: 'PROPERTY BUYING AND RENTAL', 'PROPERTY BUYING AND RENTAL Report', and 'DASHBOARD'. The user profile 'LOGESHWARAN R' is visible at the bottom left. The footer indicates 'Showing 7 of 7'.

## A12. DASHBOARD



## A13. SEARCH/FILTER OUTPUT

Home - Classroom x CSA1506\_Property\_Buying\_Ren... 192511175.simats\_saveetha - x New Tab x + Ask Gemini x

creatorapp.zoho.in/192511175.simats\_saveetha/property-buying-and-rental/#Report:PROPERTY\_BUYING\_AND\_RENTAL\_Report

PROPERTY BUYING AND RENTAL

PROPERTY BUYING AND RENTAL Report \* Save Changes Remove Changes

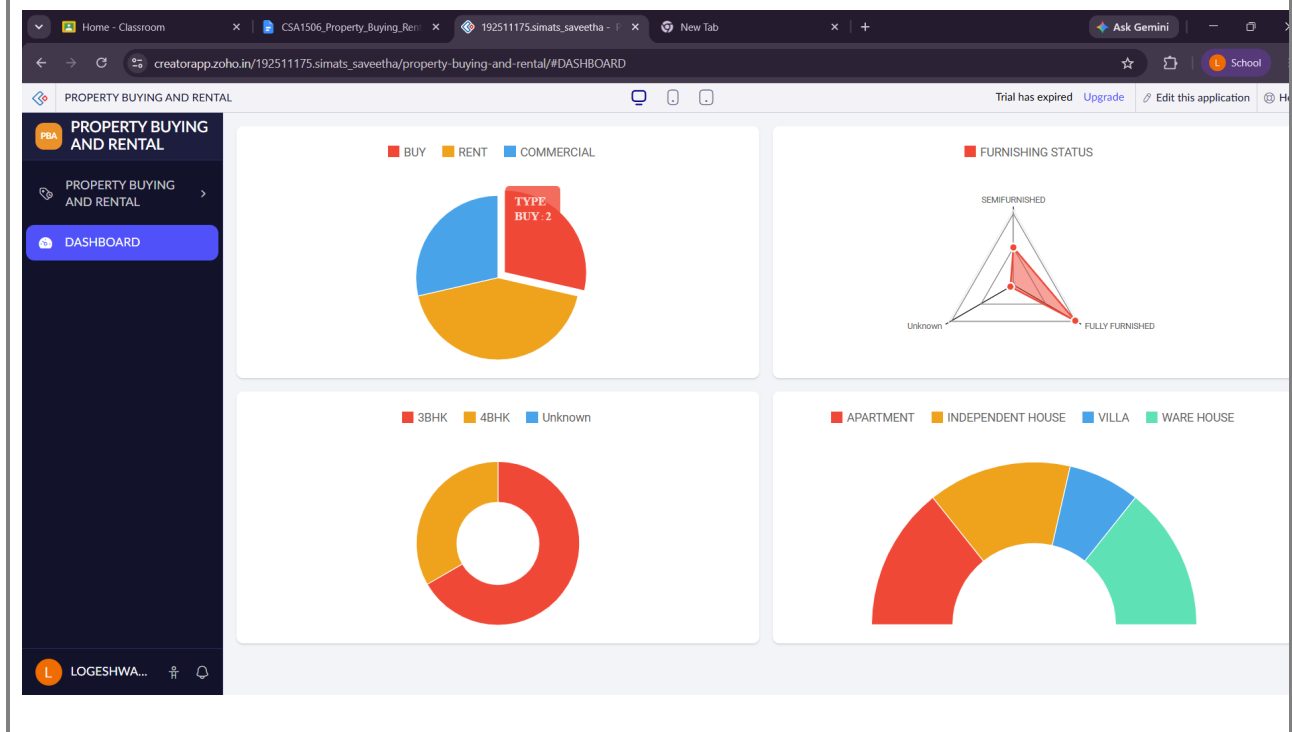
Search TYPE contains "RENT" x

| <input type="checkbox"/> | TYPE | PROPERTY TYPE     | LOCALITY    | AREA(sq.ft) | BHK/ROOMS | BUDGET RANGE | PROPERTY VALUE   |
|--------------------------|------|-------------------|-------------|-------------|-----------|--------------|------------------|
| <input type="checkbox"/> | RENT | WARE HOUSE        | TAMBARAM    | 1036        | 3BHK      | 50-75L       | ₹ 1,50,00,000.00 |
| <input type="checkbox"/> | RENT | WARE HOUSE        | KILAMBAKKAM | 3023        | 4BHK      | 1Cr+         | ₹ 1,50,00,000.00 |
| <input type="checkbox"/> | RENT | INDEPENDENT HOUSE | ADAYAR      | 4567        | 3BHK      | 75L-1Cr      | ₹ 1,50,00,000.00 |

Showing 3 of 3

LOGESHWAR...

#### A14. FINAL APPLICATION VIEW

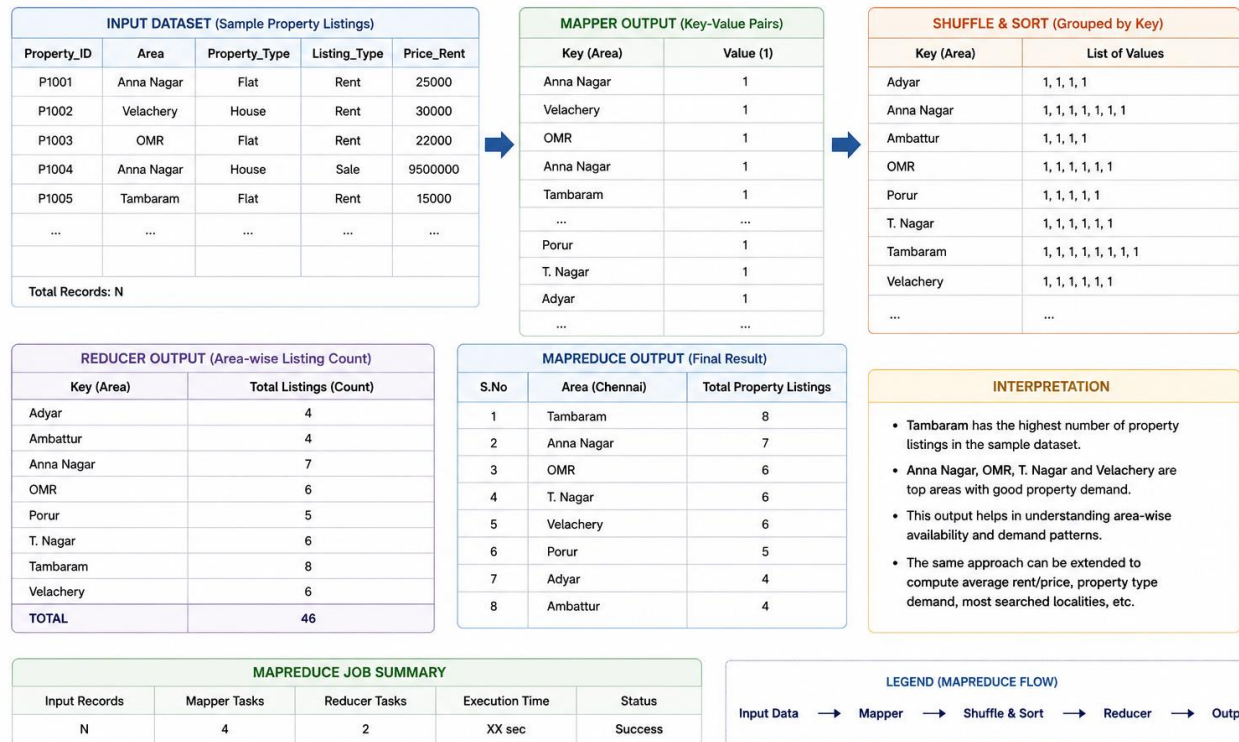




## A17. MAPREDUCE EXECUTION OUTPUT

**FIGURE 19.2 – MAPREDUCE OUTPUT**

Analytics: Area-wise Property Listing Count in Chennai



## APPENDIX B – SAMPLE DATASET FORMAT

Recommended CSV-style fields:

PropertyID,ListingType,PropertyCategory,PropertyType,PropertyTitle,OwnerAgent,Contact,Locality,Address,Landmark,BHK,BuiltUpArea,CarpetArea,Furnishing,Amenities,Availability,BudgetRange,PropertyValue,MonthlyRent,MaintenanceCharge,AdvanceDeposit,AgreementDuration,AgreementStartDate,AgreementEndDate,RegistrationStatus,PropertyStatus

The dataset should be cleaned before analytics. Recommended preparation steps are: standardize locality names, remove duplicate Property IDs, handle missing numeric values, validate

BUY/RENT-specific fields, normalize property-type names and ensure that sensitive contact information is not unnecessarily exposed in analytics outputs.

For MapReduce testing, include repeated values for Chennai localities such as Ambattur, Anna Nagar, Porur, Velachery, Tambaram, OMR, T Nagar and Adyar so that grouping and aggregation produce meaningful results.